

설계 과제 #2 테스트 프로그램 사용 방법

○ 각 테스트 프로그램 설명

- debug_test (이미 소스 코드 제공)
 - ✓ 디버그 출력은 이후 syscall_test 및 scheduler 테스트에 필요한 디버깅이 제대로 구현되었는지 확인하는 테스트
 - ✓ 이 테스트를 통과해야 다음 테스트를 진행하며, 테스트 실패 시 0점 처리
 - ✓ 자식 프로세스를 생성하여 정해진 횟수만큼 반복문을 통해 연산을 진행하며 부모는 자식을 기다리다 종료
 - ✓ debug_test의 부모는 셸이므로, debug_test 자체는 디버깅의 대상이 아니며 자식 프로세스만 디버깅 되어야 함
- syscall_test
 - ✓ 시스템 콜이 제대로 구현되었는지 확인하는 테스트
 - ✓ 이 테스트를 통과해야 scheduler_test를 진행하며 테스트 실패 시 1번만 구현된 것으로 판단
 - ✓ debug_test와 비슷하게 부모는 wait, 자식은 setticket 시스템 콜을 호출하며 종료
 - ✓ 따라서 이 테스트 또한 자식 프로세스만 디버깅 되며, 총 6틱 동안 실행되고 종료, pass 및 pid를 제외한 나머지 출력은 동일해야 함
- scheduler_test
 - ✓ 스케줄러가 제대로 구현되었는지 확인하는 테스트
 - ✓ 총 5개의 테스트로 이루어져 있으며 모두 통과해야 구현한 것으로 판단
 - ✓ 인자 없이 실행하면 모든 테스트 실행, 첫 번째 인자로 1~5 범위의 수를 입력하면 해당 테스트 실행
 - ✓ 각각의 테스트는 다양한 프로세스 개수, end_ticks, tickets 값을 가짐

○ 과제의 내용을 아무 것도 구현 하지 않은 상태에서 테스트

- 유저 프로세스가 시스템 콜을 호출하는 코드를 직접 컴파일 하지 않고, 바로 이진 파일로 추가하기 때문에 시스템 콜을 구현하지 않더라도 xv6 실행 가능함. 제대로 구현되지 않은 시스템 콜을 호출하기 때문에 xv6가 종료될 수 있음

- debug_test

```
$ debug_test
result : 1540746712
```

- syscall_test

```
$ syscall_test
6 syscall_test: unknown sys call 22
```

- scheduler_test

```
$ scheduler_test
test1 start

4 scheduler_test: unknown sys call 22
settickets error!
5 scheduler_test: unknown sys call 22
settickets error!
6 scheduler_test: unknown sys call 22
settickets error!

test1 end

test2 start

7 scheduler_test: unknown sys call 22
settickets error!
8 scheduler_test: unknown sys call 22
settickets error!
9 scheduler_test: unknown sys call 22
settickets error!

test2 end

test3 start
```

```

10 scheduler_test: unknown sys call 22
settickets error!
11 scheduler_test: unknown sys call 22
set12 scheduler_test: unknown sys call 22
settickets error!
13 scheduler_test: unknown sys call 22
settickets error!
14 scheduler_test: unknown sys call 22
settickets error!
15 scheduler_test: unknown sys call 22
tickets error!
settickets error!

test3 end

test4 start

16 scheduler_test: unknown sys call 22
settickets error!
17 scheduler_test: unknown sys call 22
settickets error!18 scheduler_test: unknown sys call 22
!
settickets error!

test4 end

test5 start

19 scheduler_test: unknown sys call 22
settickets error!
20 scheduler_test: unknown sys call 22
settickets error!
21 scheduler_test: unknown sys call 22
22 scheduler_test: unknown sys call 22
settickets error!
s23 scheduler_test: unknown sys call 22
settickets error!
24 scheduler_test: unknown sys call 22
settickets error!
ettickets error!

test5 end

```

○ 기본 설정만 구현한 상태

- debug_test (중간 출력 생략, debug_test이므로 틱 및 다른 변수 값은 신경쓰지 않음, debug 출력 형식과 동일한지만 확인)

```

$ debug_test
Process 4 start
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
result : 1540746712
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 exit

```

- syscall_test (이후 출력 생략, 해당 출력 무한 반복, debug 구현에 따라 출력이 다를 수 있지만 unknown sys call 22는 동일)

```

$ syscall_test
Process 4 start
4 syscall_test: unknown sys call 22
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)

```

- scheduler_test (중간 출력 생략, 이전 상황과 비슷함, debug 구현에 따라 출력이 다를 수 있음)

```

$ scheduler_test
test1 start

Process 4 start
Process 5 start
Process 6 start
4 scheduler_test: unknown sys call 22
settickets error!
Process 4 exit
5 scheduler_test: unknown sys call 22
settickets error!
Process 5 exit
6 scheduler_test: unknown sys call 22
Process 6 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
settickets error!
Process 6 exit

test1 end

test2 start

Process 7 start
Process 8 start
Process 9 start
7 scheduler_test: unknown sys call 22
settickets error!
Process 7 exit
Process 8 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
9 scheduler_test: unknown sys call 22
settickets error!
Process 9 exit
8 scheduler_test: unknown sys call 22
settickets error!
Process 8 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 8 exit

test2 end

test3 start

```

```

test4 end

test5 start

Process 19 start
Process 20 start
Process 21 start
19 scheduler_test: unknown sys call 22
settickets error!
Process 19 exit
20 scheduler_test: unknown sys call 22
Process 20 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
21 scheduler_test: unknown sys call 22
settickets error!
Process 21 exit
Process 22 start
Process 23 start
22 scheduler_test: unknown sys call 22
settickets error!
Process 22 exit
settickets error!
Process 20 exit
23 scheduler_test: unknown sys call 22
settickProcess 23 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
Process 24 start
24 scheduler_test: unknown sys call 22
ets error!
Process 23 exit
settickets eProcess 24 selected, stride : 0, ticket : 1, pass : 0 -> 0 (0/-1)
rror!
Process 24 exit

test5 end

```

○ 시스템 콜까지 구현한 상태

- debug_test (중간 출력 생략, debug_test이므로 틱 및 다른 변수 값은 신경쓰지 않음, debug 출력 형식과 동일하지만 확인)

```

$ debug_test
Process 4 start
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (1/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (2/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (3/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (4/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (5/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (6/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (7/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (8/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (9/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (10/-1)

Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (56/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (57/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (58/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (59/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (60/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (61/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (62/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (63/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (64/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (65/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (66/-1)
result : 1540746712
Process 4 exit

```

- syscall_test (pass 및 pid를 제외한 다른 값은 같아야 함)

```
$ syscall_test
Process 4 start
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (1/6)
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (2/6)
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (3/6)
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (4/6)
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (5/6)
Process 4 selected, stride : 1000, ticket : 100, pass : -1000 -> 0 (6/6)
Process 4 exit
```

- scheduler_test (중간 출력 생략, 스케줄러 구현이 되어있지 않아 모든 프로세스가 동일한 순서로 실행됨)

```
$ scheduler_test
test1 start

Process 4 start
Process 5 start
Process 6 start
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (1/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (1/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (1/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (2/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (2/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (2/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (3/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (3/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (3/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (4/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (4/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (4/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (5/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (5/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (5/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (6/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (6/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (6/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (7/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (7/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (7/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (8/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (8/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (8/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (9/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (9/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (9/20)
Process 4 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (10/20)
Process 5 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (10/20)
Process 6 selected, stride : 100, ticket : 1000, pass : -100 -> 0 (10/20)
```

```

Process 22 selected, stride : 7142, ticket : 14, pass : -7142 -> 0 (6/10)
Process 24 selected, stride : 1298, ticket : 77, pass : -1298 -> 0 (6/10)
Process 23 selected, stride : 1639, ticket : 61, pass : -1639 -> 0 (6/10)
Process 22 selected, stride : 7142, ticket : 14, pass : -7142 -> 0 (7/10)
Process 24 selected, stride : 1298, ticket : 77, pass : -1298 -> 0 (7/10)
Process 23 selected, stride : 1639, ticket : 61, pass : -1639 -> 0 (7/10)
Process 22 selected, stride : 7142, ticket : 14, pass : -7142 -> 0 (8/10)
Process 24 selected, stride : 1298, ticket : 77, pass : -1298 -> 0 (8/10)
Process 23 selected, stride : 1639, ticket : 61, pass : -1639 -> 0 (8/10)
Process 22 selected, stride : 7142, ticket : 14, pass : -7142 -> 0 (9/10)
Process 24 selected, stride : 1298, ticket : 77, pass : -1298 -> 0 (9/10)
Process 23 selected, stride : 1639, ticket : 61, pass : -1639 -> 0 (9/10)
Process 22 selected, stride : 7142, ticket : 14, pass : -7142 -> 0 (10/10)
Process 22 exit
Process 24 selected, stride : 1298, ticket : 77, pass : -1298 -> 0 (10/10)
Process 24 exit
Process 23 selected, stride : 1639, ticket : 61, pass : -1639 -> 0 (10/10)
Process 23 exit

test5 end

```

- 필수 구현을 모두 구현한 상태
- debug_test (중간 출력 생략)

```

$ debug_test
Process 4 start
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (1/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (2/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (3/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (4/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (5/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (6/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (7/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (8/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (9/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (10/-1)

Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (53/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (54/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (55/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (56/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (57/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (58/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (59/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (60/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (61/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (62/-1)
Process 4 selected, stride : 0, ticket : 1, pass : 0 -> 0 (63/-1)
result : 1540746712
Process 4 exit

```

- syscall_test

```
$ syscall_test
Process 4 start
Process 4 selected, stride : 1000, ticket : 100, pass : 0 -> 1000 (1/6)
Process 4 selected, stride : 1000, ticket : 100, pass : 1000 -> 2000 (2/6)
Process 4 selected, stride : 1000, ticket : 100, pass : 2000 -> 3000 (3/6)
Process 4 selected, stride : 1000, ticket : 100, pass : 3000 -> 4000 (4/6)
Process 4 selected, stride : 1000, ticket : 100, pass : 4000 -> 5000 (5/6)
Process 4 selected, stride : 1000, ticket : 100, pass : 5000 -> 6000 (6/6)
Process 4 exit
```

- scheduler_test (중간 출력 생략, 전체 내용은 텍스트 파일 참고)

```
$ scheduler_test
test1 start

Process 4 start
Process 5 start
Process 6 start
Process 4 selected, stride : 100, ticket : 1000, pass : 0 -> 100 (1/20)
Process 5 selected, stride : 100, ticket : 1000, pass : 0 -> 100 (1/20)
Process 6 selected, stride : 100, ticket : 1000, pass : 0 -> 100 (1/20)
Process 4 selected, stride : 100, ticket : 1000, pass : 100 -> 200 (2/20)
Process 5 selected, stride : 100, ticket : 1000, pass : 100 -> 200 (2/20)
Process 6 selected, stride : 100, ticket : 1000, pass : 100 -> 200 (2/20)

Process 24 selected, stride : 1298, ticket : 77, pass : 0 -> 1298 (7/10)
Process 24 selected, stride : 1298, ticket : 77, pass : 1298 -> 2596 (8/10)
Process 24 selected, stride : 1298, ticket : 77, pass : 2596 -> 3894 (9/10)
Process 24 selected, stride : 1298, ticket : 77, pass : 3894 -> 5192 (10/10)
Process 24 exit
Process 22 selected, stride : 7142, ticket : 14, pass : 5443 -> 12585 (4/10)
Process 19 selected, stride : 11111, ticket : 9, pass : 7500 -> 18611 (6/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 0 -> 7142 (5/10)
Process 19 selected, stride : 11111, ticket : 9, pass : 6026 -> 17137 (7/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 0 -> 7142 (6/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 7142 -> 14284 (7/10)
Process 19 selected, stride : 11111, ticket : 9, pass : 7500 -> 18611 (8/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 0 -> 7142 (8/10)
Process 19 selected, stride : 11111, ticket : 9, pass : 4327 -> 15438 (9/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 0 -> 7142 (9/10)
Process 22 selected, stride : 7142, ticket : 14, pass : 7142 -> 14284 (10/10)
Process 22 exit
Process 19 selected, stride : 11111, ticket : 9, pass : 7500 -> 18611 (10/10)
Process 19 exit

test5 end
```